

```

import pandas as pd

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier, export_text

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

data = pd.read_csv("play_tennis_dataset.csv")

data = data.dropna()

if "Day" in data.columns:

    data = data.drop("Day", axis=1)

encoders = {}

for col in data.columns:

    le = LabelEncoder()

    data[col] = le.fit_transform(data[col])

    encoders[col] = le

X = data.iloc[:, :-1]

y = data.iloc[:, -1]

model = DecisionTreeClassifier(

    criterion="entropy",

    random_state=0

)

model.fit(X, y)

y_pred = model.predict(X)

print("===== ID3 DECISION TREE =====")

print("Number of Training Samples:", len(data))

print("Features:", list(X.columns))

print("\nEncoded Dataset:")

print(data)

print("\nDecision Tree Rules:")

print(

    export_text(

```

```

model,

feature_names=list(X.columns),

max_depth=3

)

)

print("\nActual Values:")

print(y.values)


print("\nPredicted Values:")


print("\nPrediction for First Sample:")

print(
    encoders["Play"].inverse_transform(prediction)[0]
)

```

```

[5425 rows x 5 columns]

... Decision Tree Rules:
|--- Outlook <= 0.50
|   |--- Temperature <= 0.50
|   |   |--- class: 1
|   |--- Temperature > 0.50
|   |   |--- Temperature <= 1.50
|   |   |   |--- Humidity <= 0.50
|   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |--- Humidity > 0.50
|   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |--- Temperature > 1.50
|   |   |   |   |--- Wind <= 0.50
|   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |--- Wind > 0.50
|   |   |   |   |   |--- class: 1
|--- Outlook > 0.50
|   |--- Temperature <= 0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- Humidity <= 0.50
|   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |--- Humidity > 0.50
|   |   |   |   |   |--- truncated branch of depth 2
|   |   |--- Wind > 0.50

[1 1 1 ... 1 0 1]
, Predicted Values:
[1 1 1 ... 1 0 1]

Accuracy:
0.8956682027649769

Accuracy Percentage:
89.5668202764977 %

Confusion Matrix:
[[1238 566]
 [  0 3621]]

Classification Report:
              precision    recall  f1-score   support

     No         1.00        0.69        0.81       1804
     Yes         0.86        1.00        0.93       3621

 accuracy         0.90
 macro avg        0.93        0.84        0.87       5425
weighted avg        0.91        0.90        0.89       5425

```